

UNITY ESS

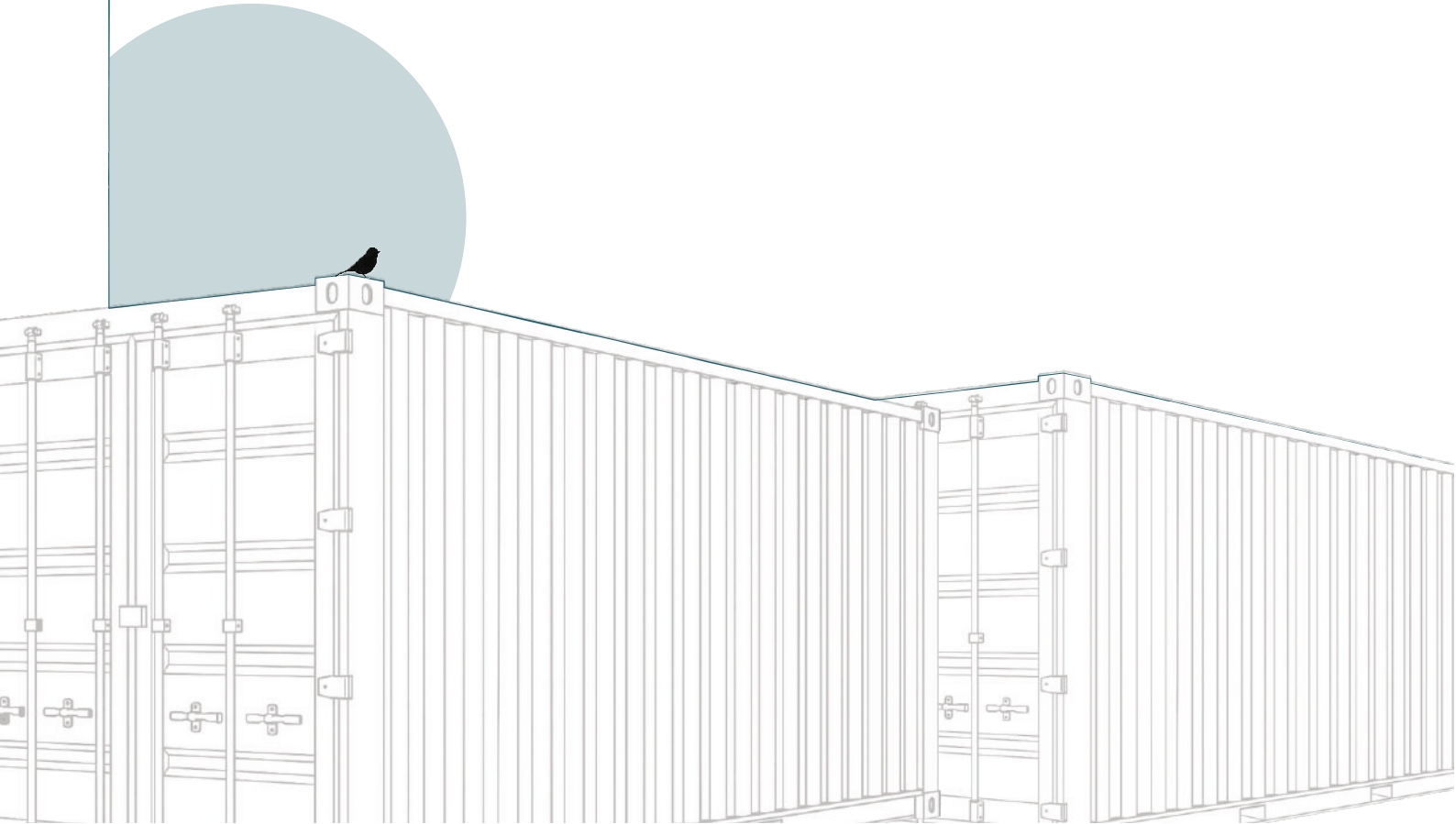
Battery Energy Storage System

Energy,
when you
need it.

CATALOGUE

2026

Energy when you need it



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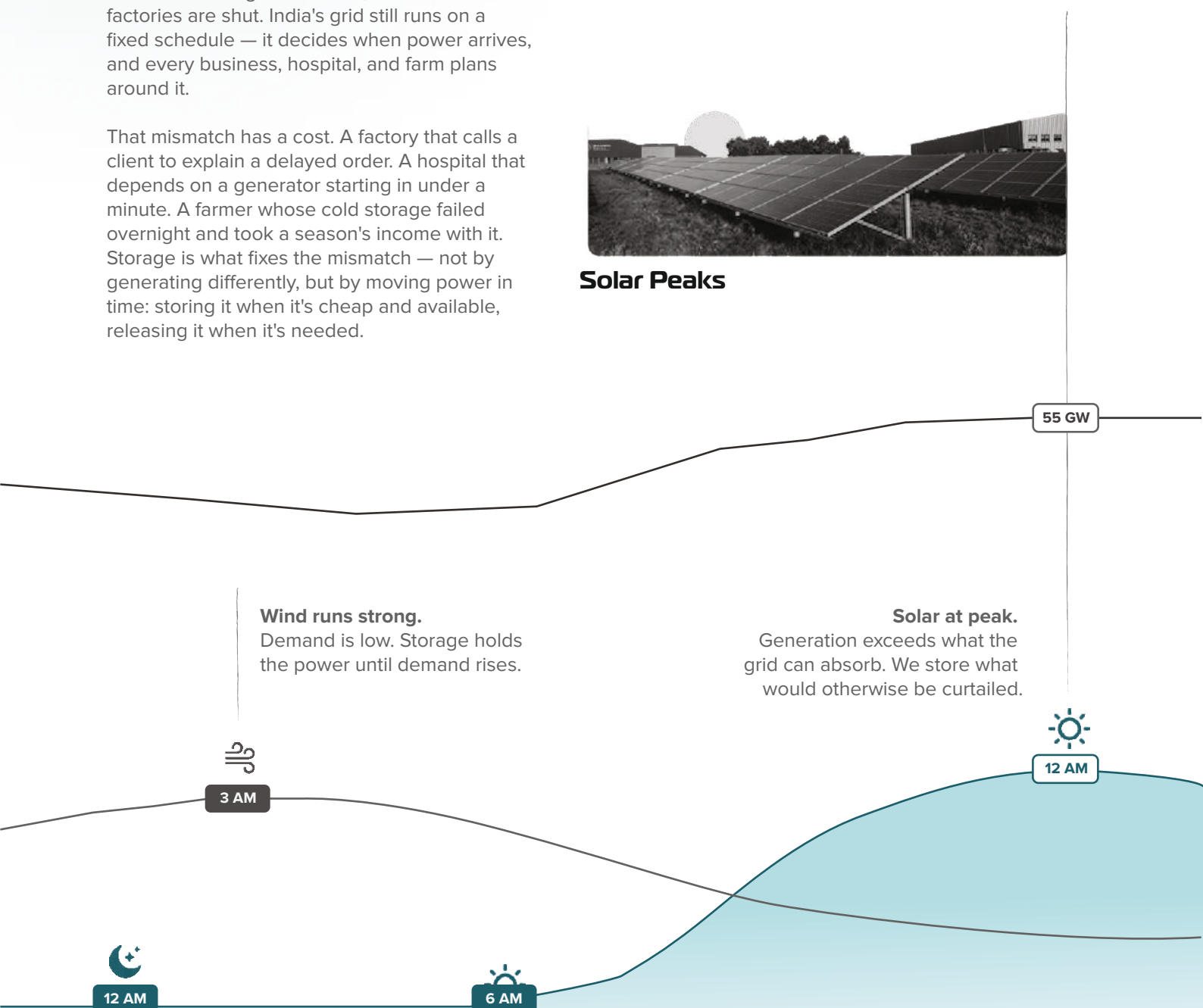
CONTENT

THE GAP WE FOUND

Solar at noon. Demand at seven. Storage connects them.

Solar peaks at noon. Demand peaks at 7 p.m. Wind runs strongest at 3 a.m., when most factories are shut. India's grid still runs on a fixed schedule — it decides when power arrives, and every business, hospital, and farm plans around it.

That mismatch has a cost. A factory that calls a client to explain a delayed order. A hospital that depends on a generator starting in under a minute. A farmer whose cold storage failed overnight and took a season's income with it. Storage is what fixes the mismatch — not by generating differently, but by moving power in time: storing it when it's cheap and available, releasing it when it's needed.

**Solar Peaks**

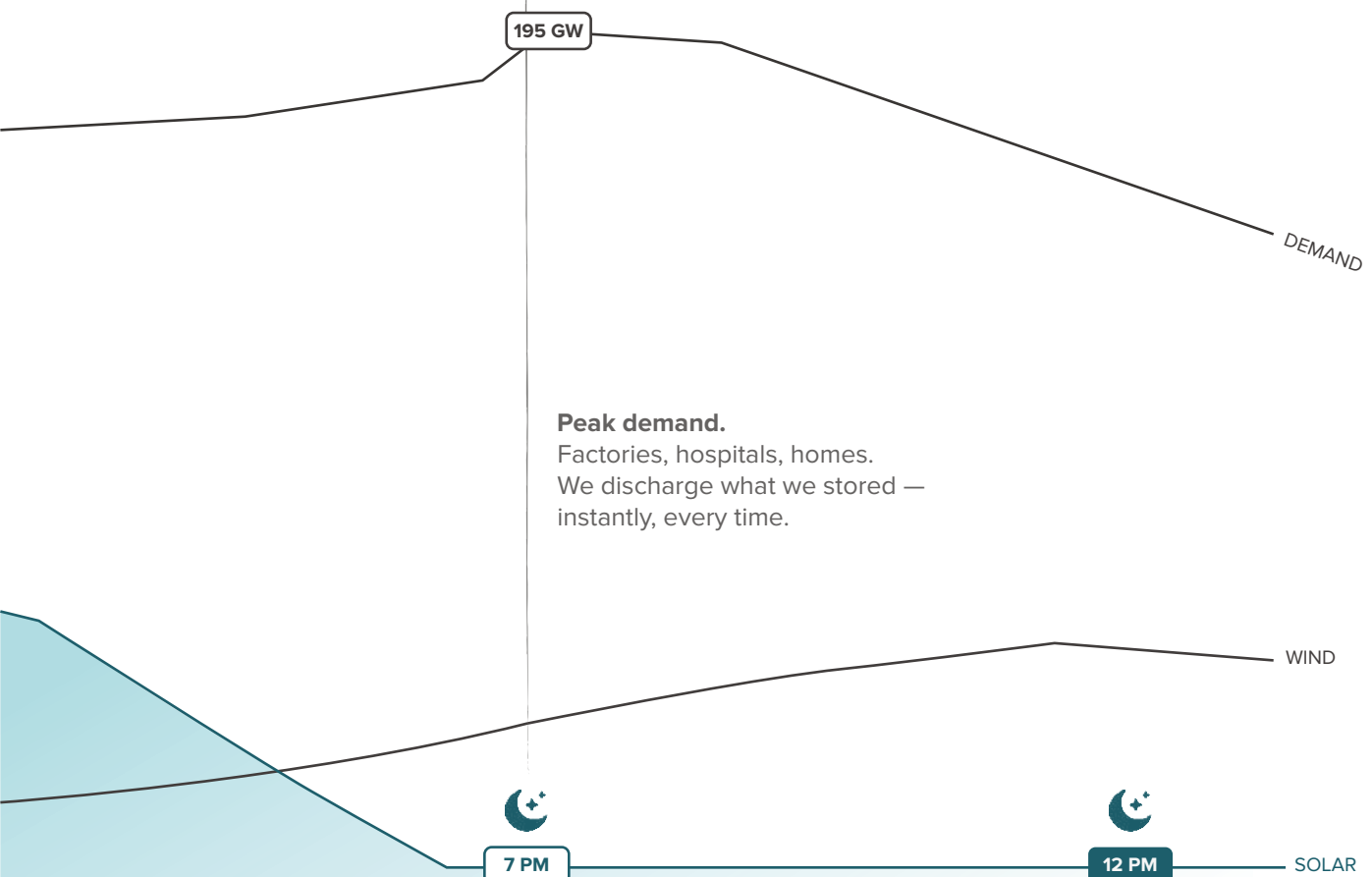
“

The difference between power that is there when you need it and power that isn't -

Unity ESS close that gap.”

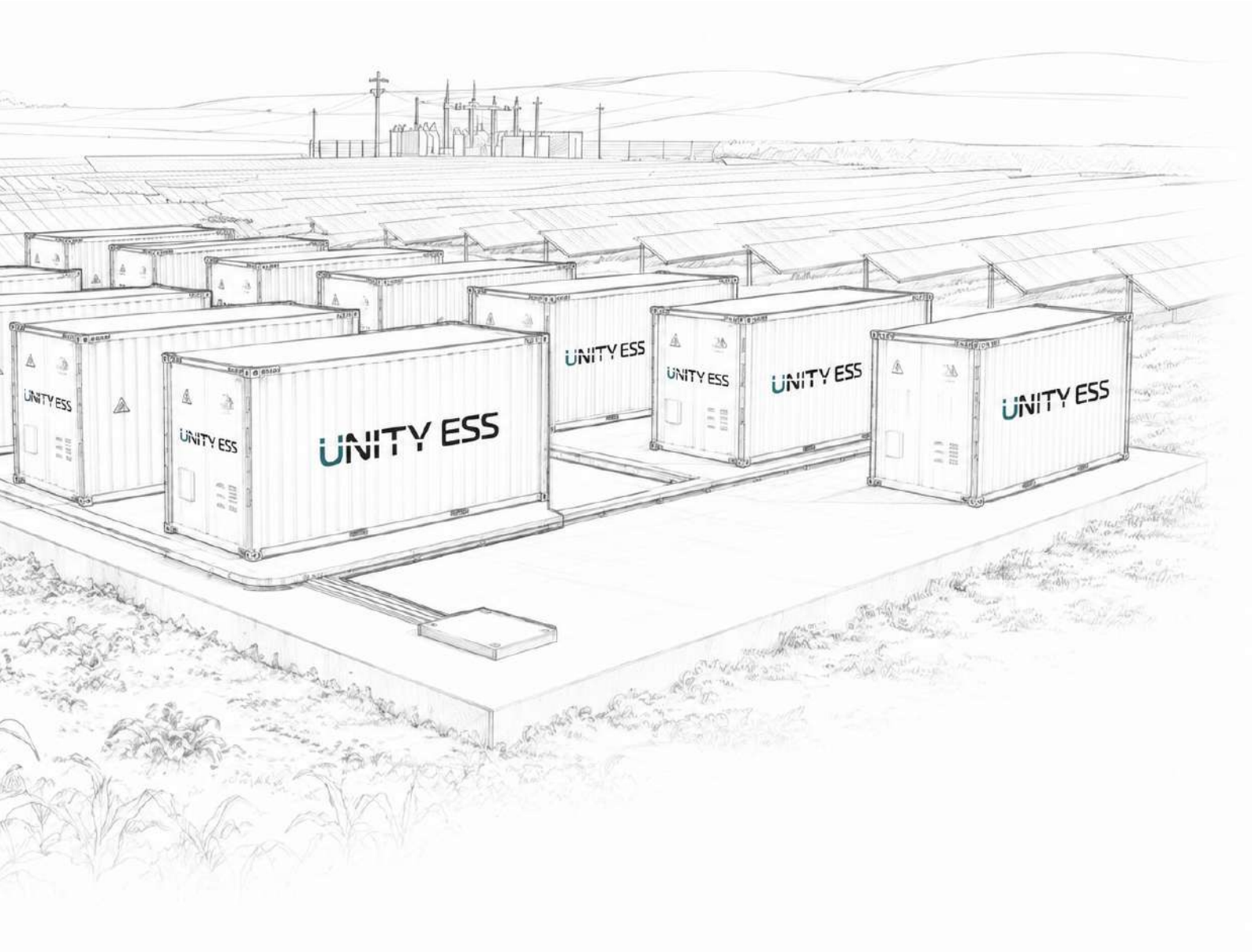


Demand Peaks



Peak demand.

Factories, hospitals, homes.
We discharge what we stored —
instantly, every time.



INTRODUCTION

What is BESS?

A Battery Energy Storage System captures energy when it's available — from solar, from the grid at off-peak rates, from wind — and releases it exactly when it's needed, on whatever schedule your operation runs on.

At its core, a BESS is battery cells, power electronics, a battery management system, thermal management, and fire safety — working as one integrated system. The hardware stores the energy. The software decides when to charge, hold, and dispatch it. Most companies treat that software as an afterthought. We don't.



WHERE SUN MEETS STORAGE

We store energy – so
when you need it,
it is there.

We are a joint venture of Ornate Solar and Inverted Energy - two companies that have spent years on opposite sides of the same problem. Generation was growing. Storage was lagging.

So we built Unity ESS to be the kind of company we wanted to work with. We do not sell boxes. We commission systems - sizing, civil drawings, electrical integration, lifecycle simulation, commissioning, monitoring, and O&M. Every system we deploy runs 24/7 monitoring from our operations centre, from commissioning day to the end of asset life.

The hardware is precise. The software is ours. The team is Indian. Built to last twenty years, not to make it past commissioning.

1 GW+

Deployed across India by Ornate Solar over a decade.

Unity EMS

Built by us, running on every system we commission.

8.5+ GW

Annual capacity. 1M+ batteries built by Inverted Energy.

25+ years

Trust in business

#Always ü



THREE SYSTEMS. ONE FAMILY.

From commercial sites to utility- scale grid infrastructure.

WARRANTY

All three systems ship with a 5-year standard warranty on battery packs, extendable to 10 years.

AIR COOLED

Unity Aura

Commercial and industrial operations. Factories, warehouses, hospitals, campuses. The system that keeps your operation running when the grid does not.

100 – 225 kWh



LIQUID COOLED

Unity Aqua

Open access applications and large industrial sites. Stores low cost renewable energy for utilisation and delivers reliable power on demand.

250 – 2,500 kWh

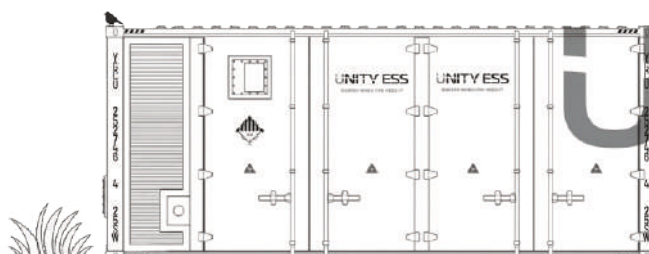


CONTAINERIZED

Unity Ultima

Utility-scale solar and hybrid projects. 20ft containerised unit with PCS, thermal management, fire protection, and advanced EMS in a single integrated system.

3.0 – 6.6 MWh



AIR COOLED • COMMERCIAL AND INDUSTRIAL

AURA



1 Product Family

QESS- Commerical series
BESS- Indusrial series
UESS- Utility series

2 Output Voltage

415V

3 Power Backup

up to 64 kW

4 Backup Time

0.5 = 2 hours backup
0.25 = 4 hours backup

5 C is Charging/ Discharging Rate

Aura

100 kWh – 225 kWh



The system

Power your operations with a BESS built for India's toughest industrial conditions. The Unity Aura delivers clean, reliable, uninterrupted power - reducing diesel dependency, cutting peak energy costs, and stabilizing your plant when the grid does not.

What makes it different?



Aura is a complete outdoor power system: cells, BMS, power conversion, thermal management, and fire protection, built into one standardised cabinet — self-contained, engineered for Indian site conditions, rated to run 20 years.



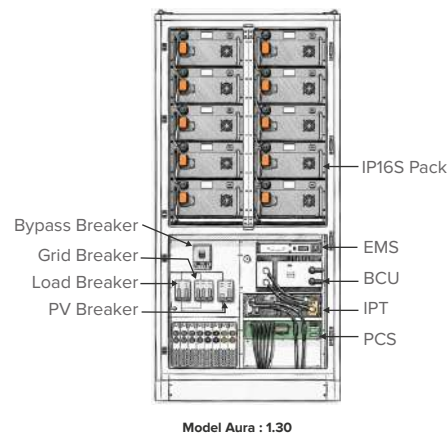
Who is it for ?

Factories, warehouses, hospitals, data centres, cold storage facilities. Any operation where a power cut costs more than the system. For behind-the-meter commercial and industrial applications.

Aura

AIR COOLED • COMMERCIAL & INDUSTRIAL

Power your operations with a BESS built for India's toughest industrial conditions. Delivers clean, reliable, and uninterrupted power — reducing diesel usage, cutting energy bills, and stabilizing your plant during peak loads



PRODUCT FEATURES



Built in MPPT for solar input



High-capacity LFP cells with 10,000 cycle life. Built and validated by Inverted Energy .



From dispatch, predictive optimization, monitoring, to tari arbitrage all managed by the Unity EMS platform.

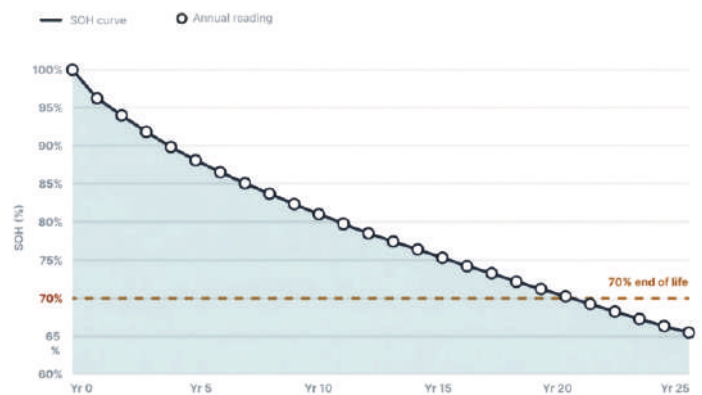


Designed for scalability from 100-225kWh by adding modules parallelly.



Built-in safety through cell level BMS to system-level protection and automatic fire protection

Battery Degradation



Product
model

Aura

Aux consumption (130 kWh)

3 kW
while operating

0.7 kW
while idle

AC side (Grid connected)

	Aura 1.0		Aura 1.30	
Rated capacity	100 kWh	100 kWh	130 kWh	130 kWh
AC rated power	25 kW	50 kW	35 kW	65 kW
Nominal Voltage (AC)	415 V			
Rated Voltage (AC)	380 - 420 Vac			
Frequency	49.5 - 50.5 Hz			
THDi	≤ 3%			
Power factor	1.0 leading - 1.0 lagging			

AC side (Off grid)

Rated output power	25 kW	50 kW	35 kW	65 kW
No. of hours of Backup	4 hours	2 hours	4 hours	2 hours
Rated output voltage	415 V			
Overload capacity	110% for 10 minutes			

PV side

Maximum PV loading	100 kW			
Maximum input current	160 A			
PV rated voltage (DC)	650 - 800 Vdc			
Input switch	250 A / 1000 Vdc / 2p, Supports single MPPT input			

Battery side

Cell	Lithium Ferrous Phosphate			
Battery pack	1P16S			
Nominal DC voltage	308 V		410 V	
DC Voltage range	269 V ~ 461 V			
Cooling method	Air cooled (HVAC)			

Distribution and Switching time

Intelligent power transfer (IPT)	144 A / 110 kW			
Switching time	< 20 ms			

General parameters

Battery efficiency	> 95%			
Communication	RS485 / Ethernet TCP IP			
Fire Fighting system	Temperature / Smoke detection, Aerosol, Flooding			
Noise	< 80 db			
IP Rating	IP54			
Dimension (W x D x H)	1050 x 1050 x 2200 mm			
Weight	1500 kgs ± 100 kgs			
Standard	IEC62619, IEC61000, UL9540(A), UN38.3			

LIQUID COOLED • OPEN ACCESS AND INDUSTRIAL

AQIJA



1

Product Family

QESS- Commercial series
BESS- Industrial series
UESS- Utility series

2

Output Voltage

4 - 415V
6 - 690V
8 - 800V

3

Power Backup

≥ 125 kW

4

Backup Time

0.5 = 2 hours backup
0.25 = 4 hours backup

5

C is Charging/
Discharging Rate

Aqua

250 kWh – 3,000 kWh



The system

The Unity Aqua is our industrial workhorse - a liquid-cooled cabinet system designed for open access applications, large industrial sites, and solar-heavy C&I facilities. It stores low-cost renewable energy and delivers reliable power on demand, across the full demand cycle.

What makes it different?



Liquid cooling is not a luxury at industrial scale — it is a requirement. Liquid cooling maintains tighter thermal control across the full cell stack, which extends cycle life, maintains performance in high ambient temperatures, and allows higher power density in the same physical footprint.



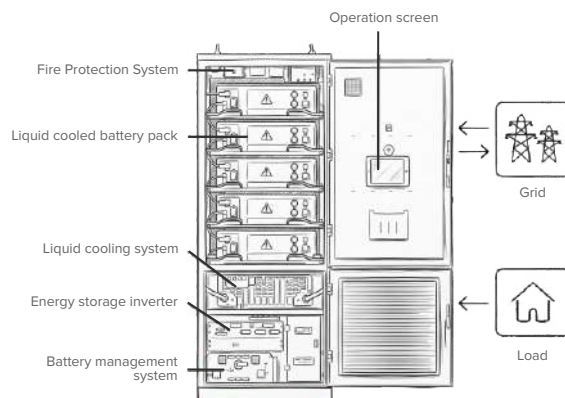
Who is it for ?

Large manufacturers, hospitals, data centres, campuses with significant energy costs. Open access electricity consumers who want to capture low-cost renewable generation and reduce grid dependence.

AQUA 4

LIQUID COOLED • BEHIND THE METER

Optimize your operations with an Energy Storage System designed for Open Access Applications. Aqua stores low-cost renewable energy for later utilization and delivers reliable power - solar when available, grid when required, storage. always balancing the two.



Model Aqua 4 : 269 kWh

PRODUCT FEATURES

Battery Degradation



Quick-connect interface between battery packs and liquid cooling system.



High-capacity LFP cells with 10,000 cycle life. Built and validated by Inverted Energy .



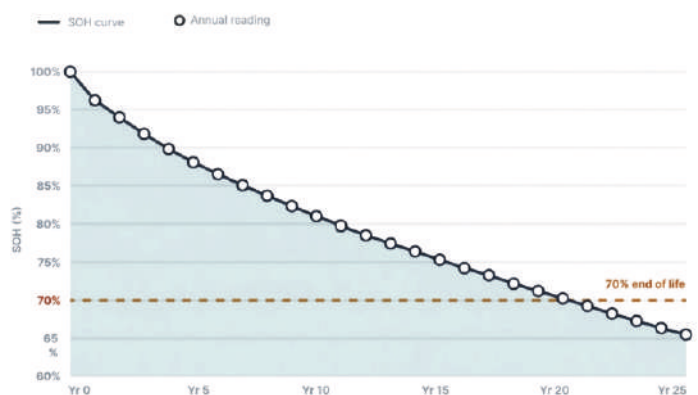
From dispatch, predictive optimization, monitoring, to tari arbitrage all managed by the Unity EMS platform.



Designed for scalability from 261 kWh to over 2.5 MWh by adding modules parallelly.



Built-in safety through cell level BMS to system-level protection and automatic fire protection



Product
model

Aqua 4

Aux consumption (269 kWh)

2.63 kW
while operating

0.05 kW
while idle

AC side

Rated capacity	269 kWh	430 kWh	539 kWh	539 kWh
AC rated power	125 kW	125 kW	125 kW	250 kW
Nominal Voltage (AC)	415 V			
Rated Voltage (AC)	380 - 420 Vac			
Frequency	49.5 - 50.5 Hz			
THDi	≤3%			
Power factor	1.0 leading - 1.0 lagging			
Overload capacity	110% for 10 minutes			

DC side

Cell	Lithium Ferrous Phosphate			
Battery pack	1P52S			
Nominal DC voltage	832 V	1331 V	832 V	832 V
DC voltage range	754 V - 936 V	1100 V - 1500 V	754 V - 936 V	754 V - 936 V
Cooling	Liquid cooled			

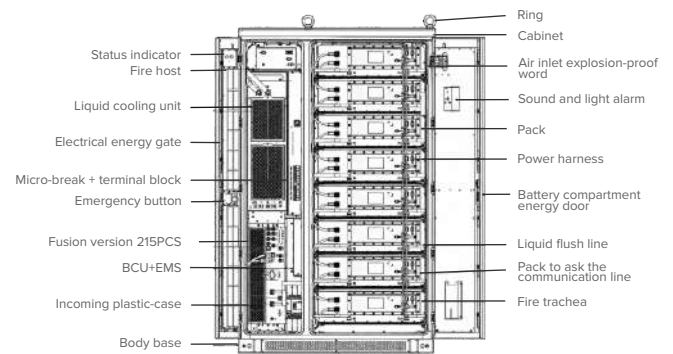
General parameters

Communication	RS485 / CAN / Ethernet			
Battery Efficiency	> 96%			
Lifetime	10,000 cycles			
Fire Fighting system	Temperature / Smoke detection, Aerosol, Flooding			
Noise	≤ 75 db			
Operating temperature	-30 to 55°C			
Max. operating altitude	3000m			
Humidity range	5% > 95% RH, No Condensation			
Corrosion level resistance	C3 - M			
IP Rating	IP54			
Dimension (W x D x H)	1100 x 1320 x 2560 mm	1710 x 1300 x 2400 mm	1920 x 1350 x 2260 mm	1920 x 1350 x 2260 mm
Weight	2600 kg ± 100 kg	3600 kg ± 100 kg	4150 kg ± 100 kg	4150 kg ± 100 kg
Standard	IEC62619, IEC61000, UL9540(A), UN38.3			

AQUA 6/8

LIQUID COOLED • OPEN ACCESS & INDUSTRIAL

Optimize your operations with an Energy Storage System designed for Open Access Applications. Aqua stores low-cost renewable energy for later utilization and delivers reliable power - solar when available, grid when required, storage. always balancing the two.



Model Aqua 6/8 : 430kWh

PRODUCT FEATURES

Battery Degradation



Quick-connect interface between battery packs and liquid cooling system.



High-capacity LFP cells with 10,000 cycle life. Built and validated by Inverted Energy .



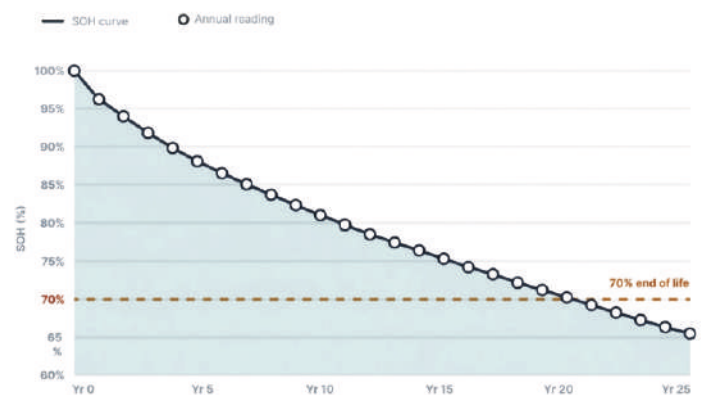
From dispatch, predictive optimization, monitoring, to tari arbitrage all managed by the Unity EMS platform.



Designed for scalability from 261 kWh to over 2.5 MWh by adding modules parallelly.



Built-in safety through cell level BMS to system-level protection and automatic fire protection



Product
model

Aqua 6/8

Aux consumption (430 kWh)

5.32 kW
while operating

0.09 kW
while idle

AC side

Rated capacity	430 kWh	860 kWh
AC rated power	215 kW	
Nominal Voltage (AC)	690 - 800V	
Rated Voltage (AC)	655 / 840 Vac	
Frequency	49.5 - 50.5 Hz	
THDi	≤ 3%	
Power factor	1.0 leading - 1.0 lagging	

DC side

Cell	Lithium Ferrous Phosphate
Battery pack	1P52S
Nominal DC voltage	1331V
DC voltage range	1100 - 1500 V
Cooling	Liquid cooled

General parameters

Communication	RS485 / CAN / Ethernet	
Battery Efficiency	> 96%	
Lifetime	10,000 cycles	
Fire Fighting system	Temperature / Smoke detection, Aerosol, Flooding	
Noise	< 80 db	
Operating temperature	-30 to 55°C	
Max. operating altitude	3000m	
Humidity range	5% > 95% RH, No Condensation	
Corrosion level resistance	C3 - M	
IP Rating	IP54	
Dimension (W x D x H)	1710 x 1400 x 2400 mm	1710 x 2800 x 2400 mm
Weight	3800 kg ± 100 kg	6600 kg ± 100 kg
Standard	IEC62619, IEC61000, UL9540(A), UN38.3	

CONTAINERIZED • UTILITY SCALE

ULTIMA



1 Product Family

QESS- Commerical series
BEES- Industrial series
UESS- Utility series

2 Output Voltage
690V

3 Power Backup
 ≥ 1250 kW

4 Backup Time
0.5 = 2 hours backup
0.25 = 4 hours backup

**5 C is Charging/
Discharging Rate**

Ultima

3.0 MWh – 6.6 MWh



The system

The Unity Ultima is built for India's fast-growing utility-scale solar and hybrid projects. A fully integrated 20ft containerized system - batteries, PCS, liquid thermal management, fire protection, and the Unity EMS - in a single deployable unit.



What makes it different?

Ultima isn't a battery you build a system around — it's the system, shipped whole. Batteries, PCS, liquid thermal management, fire protection, and Unity EMS arrive pre-integrated in one 20 ft container, from 3 MWh and scaling upward by adding containers in parallel. Most utility-scale projects lose months integrating components from different vendors before commissioning even starts. Ultima skips that.



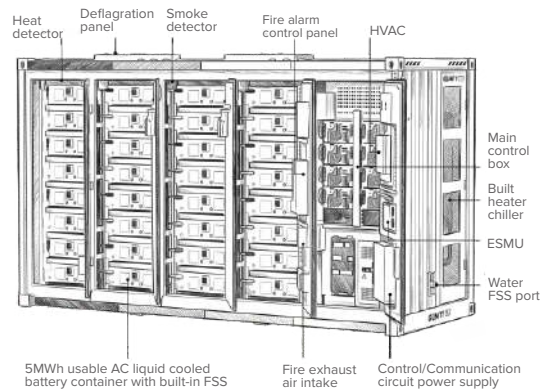
Who is it for ?

Utilities and grid operators. IPPs and renewable energy developers. Data centres. Hybrid solar plus storage projects. Any application requiring ramp-rate control, frequency response, or shifting mid-day solar generation to evening peak demand.

ULTIMA

CONTAINERISED • UTILITY SCALE

Built for India's fast-growing utility-scale solar and hybrid projects - delivering high-density, multi-megawatt-hour storage in a single containerized unit. Plug-and-operate design minimizes civil work for faster commissioning.



Model : Ultima

PRODUCT FEATURES

Battery Degradation



Suitable for coastal and high-humidity environments. IP55 protection. Built for Indian site conditions



Batteries, PCS, thermal management, fire protection, and Unity EMS in one 20ft container.



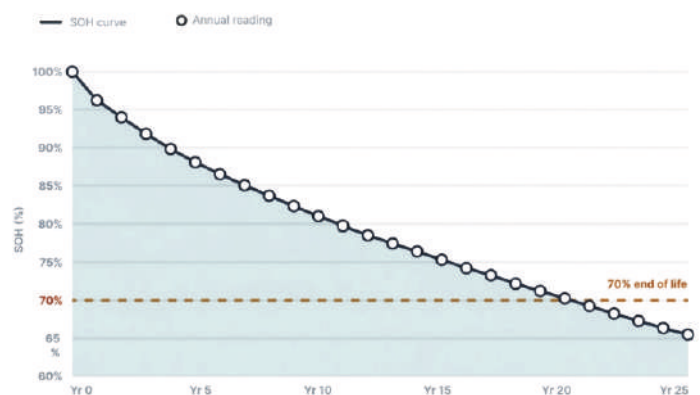
Ramp-rate control. Frequency response. Peak shaving. Mid-day solar shift to evening peak. The Unity EMS manages the full dispatch stack



High-capacity LFP cells with 10,000 cycles.



Built-in safety through cell level BMS to system-level protection and automatic fire protection



Product
model

Ultima 314

Aux consumption (5 MWh)

36.39 kW
while operating

0.71 kW
while idle

AC side

Rated capacity	3.340 MWh	4.180 MWh	5.016 MWh
AC rated power	2.5 MVA		
Nominal Voltage (AC)	690V		
Rated Voltage (AC)	655 - 724 Vac		
Frequency	49.5 - 50.5 Hz		
THDi	≤ 3%		
Power factor	1.0 leading - 1.0 lagging		

DC side

Cell	Lithium Ferrous Phosphate		
Battery pack	1P52S		
Battery rack	8P8S	10P8S	12P8S
Nominal DC voltage	1331V		
DC voltage range	1100 - 1500 V		
Cooling	Liquid cooled		

General parameters

Communication	RS485 / CAN / Ethernet		
Battery Efficiency	> 96%		
Lifetime	10,000 cycles		
Fire Fighting system	Temperature / Smoke detection, Aerosol, Flooding, alarm bell and stobe		
Noise	< 80 db		
Operating temperature	-30 to 55°C		
IP Rating	C3 - M		
Dimension (W x D x H)	6058 x 2436 x 2896 mm		
Weight	40 - 49 tons		
Standard	IEC62619, IEC61000, UL9540(A), UN38.3		

Product
model

Ultima 350

Aux consumption (5 MWh)

36.39 kW
while operating

0.71 kW
while idle

AC side

Rated capacity	3.340 MWh	4.300 MWh	5.161 MWh
AC rated power	2.5 MVA		
Nominal Voltage (AC)	690V		
Rated Voltage (AC)	655 - 724 Vac		
Frequency	49.5 - 50.5 Hz		
THDi	≤ 3%		
Power factor	1.0 leading - 1.0 lagging		

DC side

Cell	Lithium Ferrous Phosphate		
Battery pack	1P128S		
Battery rack	8P3S	10P3S	12P3S
Nominal DC voltage	1230 V		
DC voltage range	1000 - 1500 V		
Cooling	Liquid cooled		

General parameters

Communication	RS485 / CAN / Ethernet
Battery Efficiency	> 96%
Lifetime	10,000 cycles
Fire Fighting system	Temperature / Smoke detection, Aerosol, Flooding
Noise	< 80 db
Operating temperature	-30 to 55°C
IP Rating	C3 - M
Dimension (W x D x H)	6058 x 2436 x 2896 mm
Weight	40 - 49 tons
Standard	IEC62619, IEC61000, UL9540(A), UN38.3

Product
model

Ultima 587

Aux consumption (5 MWh)

36.39 kW
while operating

0.71 kW
while idle

AC side

Rated capacity	4.690 MWh	5.470 MWh	6.25 MWh
AC rated power	3.125 MVA		
Nominal Voltage (AC)	690V		
Rated Voltage (AC)	655 - 724 Vac		
Frequency	49.5 - 50.5 Hz		
THDi	≤ 3%		
Power factor	1.0 leading - 1.0 lagging		

DC side

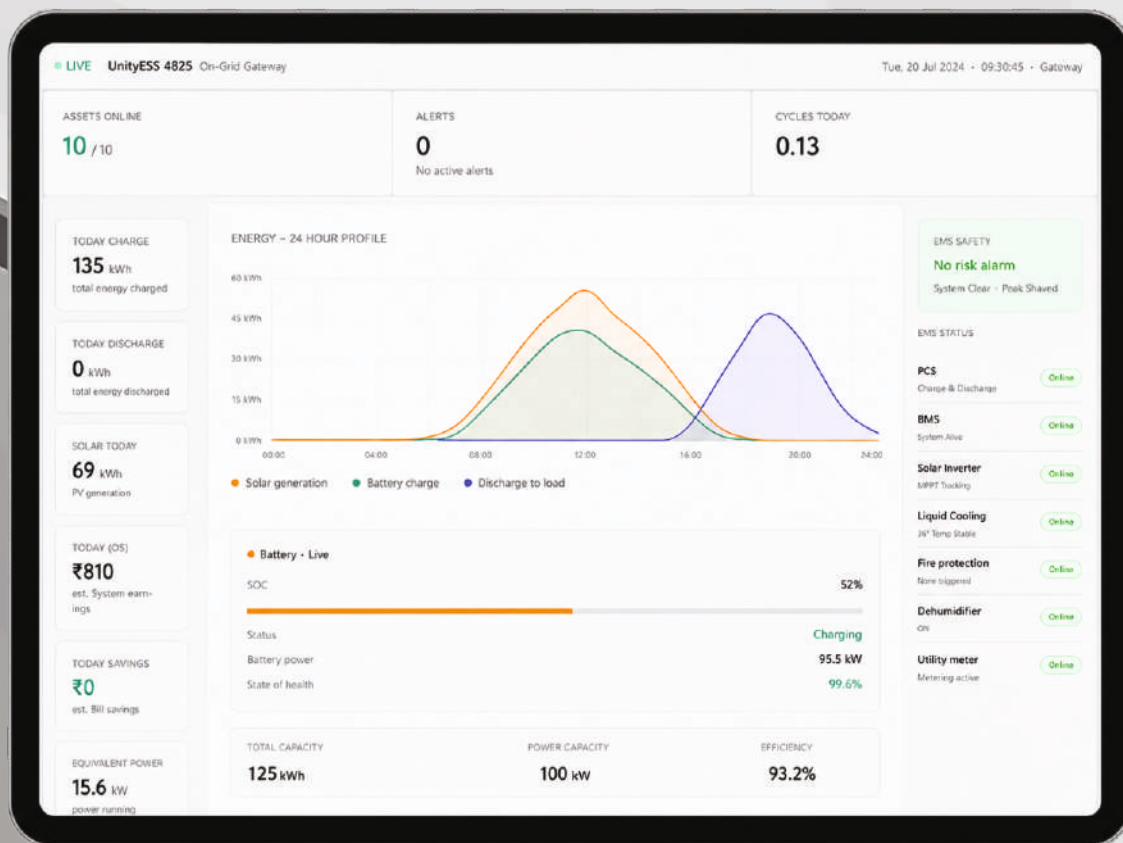
Cell	Lithium Ferrous Phosphate		
Battery pack	1P52S		
Battery rack	6P8S	7P8S	8P8S
Nominal DC voltage	1331 V		
DC voltage range	1100 - 1500 V		
Cooling	Liquid cooled		

General parameters

Communication	RS485 / CAN / Ethernet		
Battery Efficiency	> 96%		
Lifetime	10,000 cycles		
Fire Fighting system	Temperature / Smoke detection, Aerosol, Flooding, Alarm bell and strobe		
Noise	< 80 db		
Operating temperature	-30 to 55°C		
IP Rating	C3 - M		
Dimension (W x D x H)	6058 x 2436 x 2896 mm		
Weight	40 - 49 tons		
Standard	IEC62619, IEC61000, UL9540(A), UN38.3		

Built with intelligence at every layer.

The Unity EMS is what separates a battery from a storage system. Real-time dispatch, predictive optimisation, fire safety, and subsystem monitoring — all in one platform, built by us, running on every system we commission.



Find your system

Three systems, one family — sized differently for where they deploy. Match your load, power need, and site against the table below.

	Aura Commercial	Aqua Industrial	Ultima Utility
Range	100-225 kWh	261-3,000 kWh	3-6.25 kWh
Power output	50 kW-100 kW	125 kW-1,500 kW	Up to 3.125 MW
Discharge duration	2hr/4hr configs	2hr/4hr configs	2hr/4hr configs
Form factor	Outdoor/Indoor cabinet	Outdoor cabinet	Container
Cycle life	8000	≥ 10,000	≥ 10,000
Best for	Resorts, hotels, data centres, CPOs	Industries, Open Access applications	Utility-scale solar, grid frequency regulation,IPP/EPC, Data centres

Applications



Peak load shifting, demand charge reduction, and backup for factories that cannot afford to go dark — even for a minute.

Industrial



Frequency regulation, voltage support, and renewable smoothing — designed for IPPs, utilities, and grid operators managing India's fast-growing renewable capacity.

Utility Grid Operators



Instant switchover, continuous power, and surgical suite backup for hospitals that cannot tolerate a single second of interruption.

Hospitals



Capture the energy your panels generate at peak, store it, and dispatch it exactly when your load - or the grid - needs it most.

Solar + Storage hybrid



Sub-20ms switchover, UPS replacement, and peak demand management for data centres and critical IT infrastructure that operate at 100% availability.

Data centers



Cold storage backup, irrigation power, and post harvest protection — for farmers and cooperatives who cannot afford what happens when the grid goes dark.

Agriculture



Our Projects



Unity Aqua

A first-of-its-kind BESS-integrated EV charging deployment, enabling charging during grid outages and supporting peak demand.

Coimbatore,
Tamil Nadu

500 kWh



Unity Aqua

An integrated solar PV and Battery Energy Storage solution, enabling efficient energy management and reliable grid support.

Bikaner,
Rajasthan

15+ MWh



Unity Ultima

Our 5 MWh containerized BESS units came together to create a 36 MWh utility-scale energy storage deployment for a truly sustainable future.

Phalodi,
Rajasthan

36 MWh



Unity Ultima

A UnityESS deployment for Maharashtra, enabling businesses to store surplus solar energy during the day and offset high-cost grid electricity during peak tariff hours.

Karanja,
Maharashtra

3.34 MWh

We do not hand the project off after commissioning. Our same team that sized the system, drew the electrical single line, visited the site, and commissioned the hardware is the team monitoring it in year seven.

That continuity matters because storage systems don't fail at commissioning. They fail years in, when a parameter drifts or a cell ages faster than modelled. A team that knows your site catches that early. A support desk that's never seen it doesn't.

8000+

Battery/Solar Distributors
OEMs/Dealers

10*

Patents
Awarded

1000+

EPCs Satisfied
Clients

70+

Engineering
Team



UNITY ESS

Built for **You** • Always with **You** • Always **You**

Contact us

Our mission is to accelerate India's transition to clean energy through accessible, safe, and high performance storage solutions. That's why our team is focussed on providing dependable technical support and seamless customer assistance at every stage. From system setup, to guidance to after-sales assistance, we are here for you.

For sales and ordering, please contact



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